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MASTER IN ARTIFICIAL INTELLIGENCE

Head Pose Estimation under Fisheye Distortion: From Temporal Smoothing to Fisheye-Aware Adaptation

FINAL PROJECT

Course:
Computer Vision II

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1 Problem statement

Head pose estimation (HPE) estimates the 3D orientation of a human head, commonly expressed as yaw, roll and pitch. It is relevant for driver monitoring, gaze and attention analysis, human–robot interaction, surveillance and augmented reality. Modern HPE models perform well on conventional perspective images, but their robustness under fisheye or ultra-wide cameras remains an open challenge. This matters because fisheye cameras provide a much wider field of view, making them attractive for real-world monitoring scenarios, but faces near the image periphery are geometrically distorted in ways that differ from the perspective images used to train most HPE models.

The open challenge addressed in this project is therefore not simply head pose estimation in general, but head pose estimation under a structured geometric domain shift. In a standard perspective image, changes in facial appearance are mostly caused by the true 3D head orientation, illumination, identity and occlusion. In a fisheye image, the same true head pose may look different depending on where the face is located in the image. This creates a confounding factor: the model may interpret distortion-induced deformation as an actual change in head pose.

This project studies that challenge using BIWI-360, the fisheye-distorted version of the BIWI benchmark from HPE-360. In BIWI-360, the ground-truth head pose is inherited from BIWI, while the image appearance is warped according to fisheye location parameters ρ and θ. The parameter ρ describes the radial position used to generate the distortion, while θ describes the angular location. This makes the dataset especially useful for diagnosis, because the distortion process is known and can be analysed independently of the pose labels.

The research question is:

> Can a frame-by-frame 6D head pose estimator remain accurate and temporally stable on BIWI-360, and can temporal post-processing or fisheye-aware adaptation reduce the errors caused by fisheye distortion?

The initial hypothesis was that a strong pretrained full-range model would still extract useful pose information, because its 6D rotation representation and geodesic training objective are better suited to wide head rotations than plain Euler regression. However, we expected the fisheye domain shift to produce unstable frame-wise predictions, especially in peripheral frames where the image deformation is stronger. Under this hypothesis, temporal post-processing such as exponential moving average (EMA), median filtering or robust axis-specific filtering should recover a significant part of the performance by suppressing isolated prediction outliers.

A second, more ambitious hypothesis emerged from the failure analysis: if the error is mainly caused by fisheye geometry rather than random temporal noise, then the most effective solution should be domain adaptation, not only smoothing. In that case, a model fine-tuned on fisheye-distorted data and conditioned on the distortion location (ρ, θ) should outperform purely temporal post-processing. The project therefore follows a research process in which the first hypothesis is tested, partially rejected, and then refined into a stronger architecture direction.

The notebook tests this hypothesis in stages. It first evaluates pretrained 6DRepNet360 zero-shot on BIWI-360, then verifies the inference pipeline on non-fisheye AFLW2000-3D, then performs sensitivity and failure analysis, and finally tests robust temporal filters and fisheye-aware fine-tuning. This structure is important: the architecture was not selected after seeing the final numbers. Instead, the project starts from a real survey-motivated challenge, selects a model whose representation is theoretically appropriate for wide rotations, evaluates it honestly, and then revises the interpretation when the zero-shot model fails under fisheye distortion.

2 Related work

The base architectural idea comes from 6DRepNet [**hempel2022sixd**], which predicts a continuous 6D rotation representation instead of directly regressing Euler angles. The two predicted 3D vectors are converted to a valid rotation matrix through a Gram–Schmidt mapping. This design avoids Euler discontinuities and quaternion antipodal ambiguity, while avoiding the redundancy of directly predicting all 9 entries of a rotation matrix.

6DRepNet360 [**hempel2024sixdrepnet360**] extends this idea to full-range head pose estimation. It uses a 6D rotation representation, a geodesic loss on SO(3), and additional full-rotation training data from CMU Panoptic combined with 300W-LP. This makes it an appropriate zero-shot candidate for a fisheye robustness study: it is landmark-free, public, recent and designed for wide pose ranges.

Cobo et al. [**cobo2024methodology**] provide the methodological motivation for using geodesic error. They show that Euler-angle MAE can be misleading because nearby rotations may have very different Euler decompositions, especially near problematic configurations. Therefore, this project reports Euler MAE for comparability but also uses geodesic SO(3) error as the more faithful orientation metric.

Li et al. [**li2024fisheye**] introduce the fisheye HPE setting and the HPE-360 benchmark. Their work motivates the final direction of this project: if fisheye distortion is location-dependent, the model should not only smooth predictions temporally, but also adapt to fisheye images and use distortion-location information.

Other relevant alternatives include HopeNet [**ruiz2018hopenet**], which predicts binned Euler angles, WHENet [**zhou2020whenet**], which extends yaw range, and TriNet [**cao2021trinet**], which uses a rotation-matrix/vector representation. These alternatives were considered conceptually, but 6DRepNet360 was selected because its representation and loss are directly aligned with the full 3D rotation nature of the task.

3 Architecture selection

The selected starting model is the official pretrained 6DRepNet360 checkpoint:

`6DRepNet360_Full-Rotation_300W_LP+Panoptic.pth`

from the public repository:

`https://github.com/thohemp/6DRepNet360`

The choice was made for principled reasons before observing the final results. The task requires estimating a 3D orientation, not three unrelated scalar values. A model for this setting should therefore represent rotations in a way that respects the geometry of SO(3). 6DRepNet360 satisfies this because it predicts a continuous 6D representation that is mapped into a valid rotation matrix, and it is trained with a geodesic loss rather than treating yaw, roll and pitch as independent Euclidean regression targets.

This is particularly relevant for fisheye images. Under fisheye distortion, local facial appearance can be heavily warped, so the output representation should not introduce additional discontinuities or ambiguities. Euler angles are intuitive and useful for reporting results, but they are not ideal as the internal learning representation for wide rotations. Quaternions avoid gimbal lock but have antipodal ambiguity. A full 9D rotation matrix is geometrically meaningful but redundant and requires enforcing orthogonality. The 6D representation is a compromise: it is compact, continuous for learning, and can be deterministically converted into a valid rotation matrix.

The model is also landmark-free. This is important because fisheye distortion can deform facial geometry and make landmark-based pipelines fragile. A landmark-based method would first need to estimate stable 2D or 3D facial landmarks under non-perspective distortion, and errors in this first stage would propagate to pose estimation. A landmark-free appearance model is therefore a better first candidate when facial landmarks may be unreliable near the fisheye periphery.

Another reason for selecting 6DRepNet360 is that it is already designed for wide-range orientation estimation. The project does not only study frontal or near-profile faces, but a setting where unusual apparent rotations and strong deformations can occur. A narrow-range head pose estimator could fail simply because its design is not intended for broad pose coverage. 6DRepNet360 reduces that confound: if it fails on BIWI-360, the failure is more plausibly related to fisheye domain shift than to a basic inability to represent large rotations.

At least one alternative was considered and rejected. A simpler Euler-angle model such as HopeNet would be easier to interpret and historically important, but its Euler/binning formulation is less appropriate for full 3D rotations and wide pose regimes. WHENet was also relevant because it targets wide yaw estimation, but its formulation still does not address the full SO(3) geometry as directly as 6DRepNet360. A landmark-based method was rejected because it would add another failure point under fisheye deformation. Finally, a pure temporal model was not selected as the main architecture because temporal smoothing cannot change the per-frame visual representation learned by the CNN. It can suppress jitter, but it cannot teach the model how fisheye distortion changes appearance.

The initial architecture is therefore used in a zero-shot way: each BIWI-360 frame is processed independently, and the predicted rotation is converted to Euler angles only for reporting. This makes the first experiment a clean test of whether a strong perspective-trained, full-range HPE model generalises to fisheye images.

After the failure analysis, the architecture is extended in a way that directly follows the observed mechanism. The final adaptation keeps the pretrained 6DRepNet360 backbone and rotation head, and adds a small residual correction head conditioned on fisheye location features:

$$[\rho, \sin(\theta), \cos(\theta)].$$

The residual head is zero-initialised, so training starts exactly from the pretrained model and learns only a correction. This is important because the goal is not to discard the useful representation learned by 6DRepNet360, but to adapt it to a new distortion domain.

This final model is also deployable in principle. In HPE-360, ρ and θ are known because they are used to synthesise the fisheye distortion. In a real fisheye camera, equivalent information can be obtained from the face location in the image and the calibrated lens distortion model. Thus, conditioning on location is not test-label leakage; it is a way of giving the model physically meaningful information about the distortion field.

The architecture evolution therefore reflects the research process. The project begins with a theoretically justified 6D/SO(3) model, discovers that representation alone is not enough under fisheye distortion, tests temporal post-processing as a partial remedy, and finally moves toward a fisheye-aware model that adapts the learned mapping using distortion-location information.

4 Experimental setup

The main dataset is BIWI-360 from HPE-360. The notebook evaluates all BIWI-360 frames with a pretrained 6DRepNet360 model in a zero-shot setting. Frames are split by sequence, not by individual frame, to avoid temporal leakage. The validation split contains sequences $[4, 10, 16, 22]$, with 3,049 frames and 3,043 consecutive pairs. The test split contains the remaining 20 sequences, with 12,629 frames and 12,585 consecutive pairs.

The evaluation uses:

- per-axis angular MAE for yaw, roll and pitch;
- mean Euler MAE and RMSE;
- geodesic SO(3) error;
- temporal prediction jitter;
- temporal motion error between predicted and ground-truth frame-to-frame changes.

The main preprocessing follows the 6DRepNet360 inference pipeline: images are resized, centre-cropped and normalised. Since BIWI-360 already provides frame images and labels, no additional face detector is used for the main experiment.

AFLW2000-3D is used as a non-fisheye sanity check. Faces are cropped using the available 2D landmarks, then passed through the same model transform. This checks whether poor BIWI-360 performance is due to a broken inference implementation or to the fisheye domain shift.

The final adaptation experiment fine-tunes on 300W-LP-360, not on BIWI-360. This avoids identity leakage and makes the result cross-dataset: training uses fisheye-distorted 300W-LP-360 images, while final evaluation remains on the BIWI-360 test split.

5 Results

The zero-shot model performs poorly on BIWI-360. On the test split, raw 6DRepNet360 obtains 22.60 degrees mean Euler MAE and 33.19 degrees mean geodesic error. Balanced EMA smoothing improves Euler MAE to 19.94 degrees and reduces mean prediction jitter from 44.27 to 9.22 degrees per frame. However, the geodesic error slightly worsens from 33.19 to 33.55 degrees. This means that EMA reduces temporal noise and improves per-axis averages, but it does not substantially improve the true 3D orientation.

The AFLW2000-3D sanity check shows that the pipeline is correct. With the same model and angle convention, the model obtains 6.39 degrees mean MAE on all AFLW2000-3D samples and 5.88 degrees under the common

Table 1: Main BIWI-360 test results from the final notebook.

Metric	Raw	EMA-smoothed	Change
Mean Euler MAE (deg)	22.60	19.94	-2.66
Mean geodesic error (deg)	33.19	33.55	+0.36
Mean prediction jitter (deg/frame)	44.27	9.22	-35.05

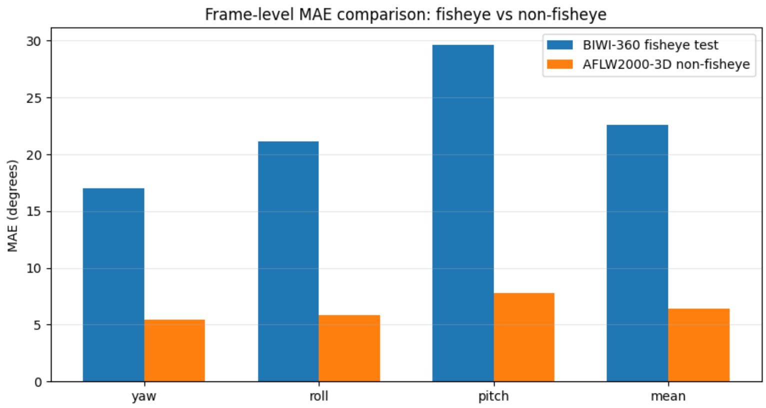

Figure 1: Sanity-check comparison between BIWI-360 and AFLW2000-3D. The same model performs much better on non-fisheye images, supporting the domain-shift interpretation.

±99° range filter. Therefore, the poor BIWI-360 result is not caused by an implementation or angle-convention error. It is mainly caused by fisheye domain shift.

The sensitivity analysis confirms that error grows with fisheye radial position. Raw mean MAE increases from 18.85 degrees for low-ρ frames to 28.35 degrees for high-ρ frames. The rate of extreme raw predictions also rises in the high-ρ group.

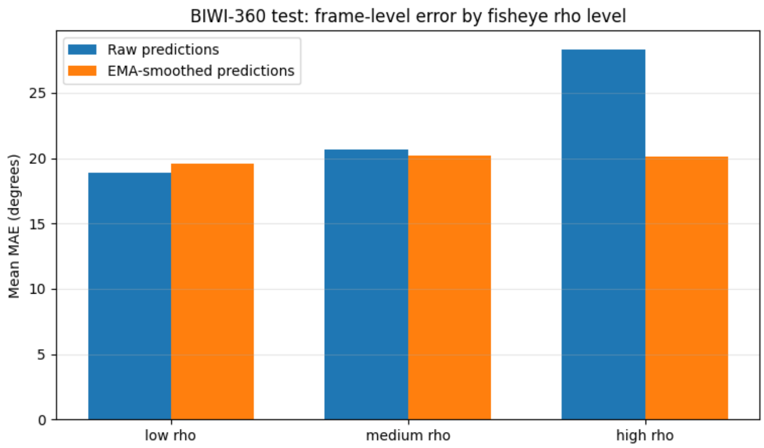

Figure 2: BIWI-360 error by fisheye radial level. Raw error is largest for high-ρ frames, indicating that stronger fisheye deformation makes zero-shot HPE harder.

The simple-baseline comparison is severe but informative. Constant predictors outperform the zero-shot model in overall mean MAE: the validation-mean baseline reaches 17.55 degrees, compared with 22.60 for raw 6DRepNet360 and 19.94 for EMA. However, the axis-specific analysis shows that the model is still useful for yaw. The global failure is mainly caused by unstable roll and pitch predictions.

Robust temporal filtering improves the accuracy–stability trade-off. The best global robust filter selected on validation is a centred median filter with window size 11. It reduces test mean MAE to 16.63 degrees and

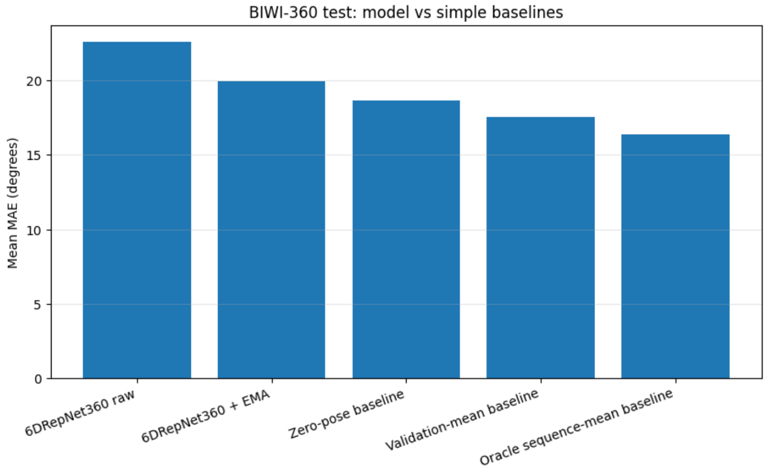

Figure 3: Simple baseline comparison. Constant predictors are competitive because roll and pitch are relatively centred in BIWI-360, while zero-shot predictions contain large outliers.

temporal motion error to 7.51 degrees per frame. An axis-specific robust strategy performs even better: centred median for yaw, and validation-set circular means for roll and pitch. This reaches 13.64 degrees mean MAE and 2.96 degrees/frame temporal motion error.

The strongest result is fisheye-aware domain adaptation. The final ablation grid in Section 16.7 shows that fine-tuning on 300W-LP-360 and conditioning on (ρ, θ) reaches 7.90 degrees mean MAE and 14.71 degrees geodesic error on the BIWI-360 test split.

Table 2: Final adaptation ablation on BIWI-360 test.

Experiment	Yaw	Roll	Pitch	Mean MAE	Geodesic
Zero-shot	16.99	21.13	29.66	22.60	33.19
Location, frozen backbone	9.35	6.35	14.79	10.16	18.49
Head only, frozen, no location	11.20	6.70	16.34	11.41	20.72
Fine-tune, no location	7.62	5.10	13.58	8.76	16.15
Fine-tune + location	7.22	4.74	11.75	7.90	14.71

This ablation revises the interpretation. Location conditioning helps, but fisheye fine-tuning is the dominant factor. Training only a residual head on frozen features already halves the error, showing that pretrained features remain useful but the original output mapping is poorly calibrated for fisheye images. Full fine-tuning improves further, and location metadata gives a smaller but consistent additional gain.

6 Failure analysis

The first failure case is severe fisheye deformation. In the worst frame-level examples from Section 11, several frames have visible geometric distortion, black warped regions or peripheral face placement. For example, the worst test frames reach mean per-frame errors around 150 degrees, with predicted roll or pitch values close to $\pm 180°$ while the ground truth remains moderate. Mechanistically, this connects to the architecture because 6DRepNet360 was trained mostly on perspective or non-fisheye images. Its CNN features and rotation head learn appearance-to-pose cues that are no longer calibrated when the face geometry is warped by fisheye distortion.

Figure 4: Worst BIWI-360 frame-level failures. These examples show that some errors are not small deviations but full orientation collapses under severe fisheye distortion.

The second failure case is temporal instability caused by isolated extreme predictions. In Section 11.1, some consecutive frames have true head-pose motion below 2 degrees, while the raw predicted pose jumps by more than 250 degrees in Euler prediction space. This is not real motion; it is model instability. Only 5.2% of test frames contain at least one raw predicted angle above 150 degrees in absolute value, but those frames have about 110.3 degrees mean MAE, compared with about 17.8 degrees for frames without extreme raw predictions.

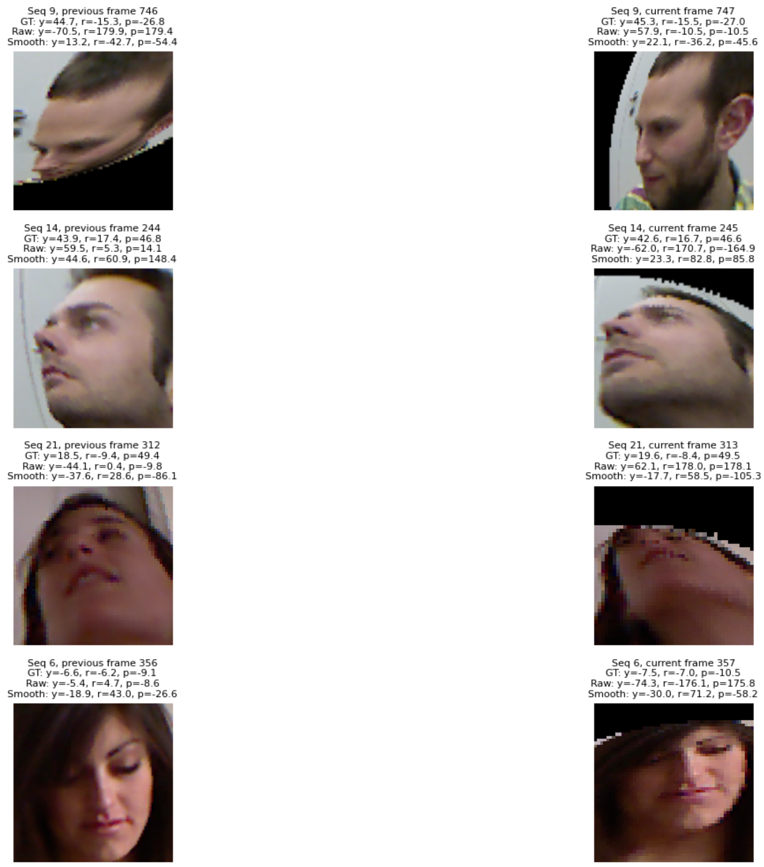

Figure 5: Worst temporal jitter events. The true pose changes smoothly, but the raw model prediction can jump by more than 250 degrees, especially in roll and pitch.

These failures explain why EMA helps but is insufficient. EMA smooths all predictions equally; it can reduce the sharpness of a jump, but it cannot know whether a raw prediction is an outlier or a valid rapid motion. It can also drag the smoothed trajectory away from the correct pose for several frames. Robust median filtering improves this behaviour, but it still cannot solve the underlying per-frame domain shift.

The baseline and axis-specific analyses reveal another important failure: the model is not uniformly bad. It captures useful yaw information, but roll and pitch are much less reliable. Since BIWI-360 roll and pitch distributions are relatively centred, constant predictors can beat the raw model in overall mean MAE. This is why the axis-specific robust filter works well diagnostically: it preserves yaw dynamics and suppresses unstable roll/pitch predictions. However, replacing two axes with constants is not a general HPE solution.

7 Thesis direction

The results partially refute the initial hypothesis. The model is indeed temporally unstable on BIWI-360, and temporal post-processing reduces this instability. However, the main cause is not temporal. The core problem is per-frame geometric domain shift caused by fisheye distortion.

This conclusion is supported by four findings. First, the same model works well on AFLW2000-3D, so the pipeline is not broken. Second, error increases with ρ, so stronger fisheye distortion makes the task harder. Third, EMA improves Euler MAE and jitter but not geodesic orientation error. Fourth, fisheye fine-tuning produces a much larger improvement than any temporal filter.

The next concrete step should therefore be a multi-seed fisheye-domain adaptation experiment. The current final ablation shows a clear ranking, but the additional benefit of location conditioning is around one degree, so it should be confirmed statistically. The proposed next experiment is:

1. repeat the four adaptation variants from Section 16.7 over at least three random seeds;
2. report mean and standard deviation for Euler MAE and geodesic error;
3. evaluate not only on BIWI-360 but also on AFLW2000-360 if available;
4. compare explicit $(\rho, \sin\theta, \cos\theta)$ conditioning with image-coordinate conditioning and with a distortion-aware spatial encoding.

The feasible thesis direction is therefore not simply “apply stronger smoothing”. A better model should combine a 6D/SO(3) rotation head, fisheye-domain fine-tuning, explicit location or distortion conditioning, and possibly uncertainty-aware outlier rejection before temporal smoothing. Temporal post-processing should remain a secondary stabilisation layer, not the main solution.

Bibliography

[1] K. Khan, R. U. Khan, R. Leonardi, P. Migliorati, and S. Benini. Head pose estimation: A survey of the last ten years. *Signal Processing: Image Communication*, 99:116479, 2021.

[2] A. Asperti and D. Filippini. Deep learning for head pose estimation: A survey. *SN Computer Science*, 4, 2023.

[3] T. Hempel, A. A. Abdelrahman, and A. Al-Hamadi. 6D rotation representation for unconstrained head pose estimation. In *IEEE International Conference on Image Processing (ICIP)*, pp. 2496–2500, 2022.

[4] T. Hempel, A. A. Abdelrahman, and A. Al-Hamadi. Toward robust and unconstrained full range of rotation head pose estimation. *IEEE Transactions on Image Processing*, 33:2377–2387, 2024.

[5] A. Cobo, R. Valle, J. M. Buenaposada, and L. Baumela. On the representation and methodology for wide and short range head pose estimation. *Pattern Recognition*, 149:110263, 2024.

[6] B. Li, D. Zhang, C. Huang, Y. Xian, M. Li, and D.-J. Lee. Location-guided head pose estimation for fisheye image. *arXiv:2402.18320*, 2024.

[7] Y. Zhou, C. Barnes, J. Lu, J. Yang, and H. Li. On the continuity of rotation representations in neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5738–5746, 2019.

[8] N. Ruiz, E. Chong, and J. M. Rehg. Fine-grained head pose estimation without keypoints. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2018.

[9] Y. Zhou and J. Gregson. WHENet: Real-time fine-grained estimation for wide range head pose. In *British Machine Vision Conference (BMVC)*, 2020.

[10] Z. Cao, Z. Chu, D. Liu, and Y. Chen. A vector-based representation to enhance head pose estimation. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, pp. 1188–1197, 2021.

[11] X. Zhu, Z. Lei, X. Liu, H. Shi, and S. Z. Li. Face alignment across large poses: A 3D solution. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 146–155, 2016.

[12] G. Fanelli, M. Dantone, J. Gall, A. Fossati, and L. Van Gool. Random forests for real time 3D face analysis. *International Journal of Computer Vision*, 101(3):437–458, 2013.